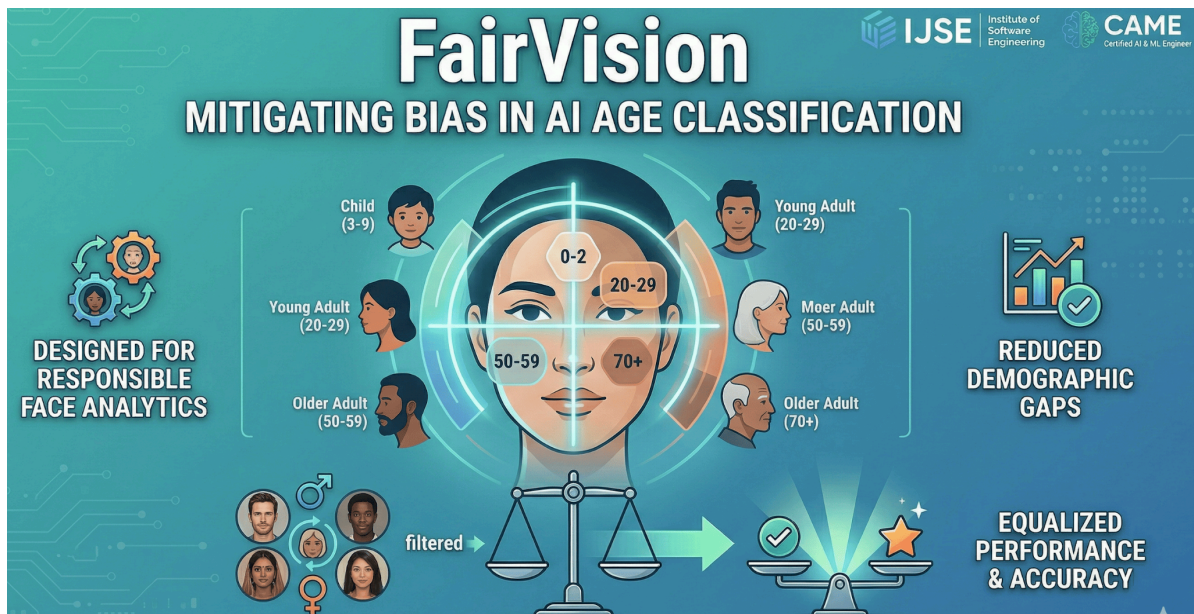


Institute of Software Engineering (IJSE)

Certified AI & ML Engineer

Individual Assignment Brief

Project Title: FairVision - Detecting and Mitigating Bias in a CNN-Based Age Group Classification System Using FairFace



Assignment Information

- **Faculty / School:** School of Computing
- **Programme:** Certified AI & ML Engineer
- **Lecturer:** Dasun Athukorala
- **Academic Year:** 2025/2026
- **Project Title:** FairVision - Detecting and Mitigating Bias in a CNN-Based Age Group Classification System Using FairFace
- **Assignment Type:** Individual Assignment
- **Assignment Weighting:** 50%
- **Total Marks:** 100
- **Date Issued:** 11/04/2026
- **Submission Deadline:** 10 May 2026, 11:00 PM
- **Submission Method:** Via Google Drive. The lecturer will share the submission link separately before the deadline.
- **Required Submission Format:** See Section 14 for details

1. Background and Industry Context

Face-based AI systems are increasingly used in security, attendance tracking, mobile authentication, smart retail, surveillance, and public services. Although these systems can achieve strong overall accuracy, they may also perform unevenly across demographic groups. Such performance gaps can create technical, ethical, and social risks.

In this project, you will work as an AI/ML engineer tasked with auditing a **convolution neural network (CNN)**-based face analytics prototype before deployment. Your responsibility is not only to build a working CNN model, but also to investigate whether the system behaves unfairly across different groups and to test practical strategies for reducing that bias.

This project is intentionally designed to go beyond a standard image-classification assignment. You are expected to combine deep learning, evaluation, responsible AI thinking, and engineering judgment.

2. Project Scenario

You are an AI/ML engineer at a company developing a face-based analytics solution for a controlled environment. Before the prototype can be approved, the company wants evidence that the system:

- works reliably on the chosen task,
- has been evaluated across different demographic groups,
- shows measurable evidence of bias detection, and
- includes mitigation strategies to improve fairness without severely harming overall performance.

You must design, implement, evaluate, and document a CNN-based solution that addresses these requirements.

3. Project Goal

Build a **CNN-based age group classification system using the FairFace dataset**, evaluate its performance across race and gender groups, identify potential bias, and **apply mitigation strategies to reduce unfairness while preserving acceptable model quality**.

4. Scope Definition

For this assignment, the phrase "face analytics prototype" refers to a controlled CNN-based face image classification system built on a fixed public dataset. This assignment uses one fixed task configuration:

- Primary dataset: FairFace (explained in the Section 6)

- Prediction task: age group classification
- Input: a single face image from the FairFace dataset
- Output: one age group label from the FairFace age categories
- Fairness audit groups: race and gender

The main focus of the project is bias detection and mitigation in a CNN-based vision system.

You are not required to build:

- a production-grade surveillance system,
- an identity-recognition system,
- a real-time deployment pipeline, or
- a legally deployable biometric product.

5. Learning Outcomes

By completing this project, students should be able to:

- design and train a CNN for age group prediction from facial images,
- prepare and analyze image datasets for supervised learning,
- evaluate model performance using standard classification metrics,
- measure subgroup performance differences and reason about fairness,
- apply and compare practical bias-mitigation techniques, and
- communicate technical findings in a professional engineering format.

6. FairFace Dataset Overview

This assignment uses the **FairFace dataset** provided through Hugging Face at [HuggingFaceM4/FairFace](https://huggingface.co/datasets/HuggingFaceM4/FairFace).

Hugging Face Link for the dataset: <https://huggingface.co/datasets/HuggingFaceM4/FairFace>

FairFace is a public face attribute dataset created to support more balanced evaluation across demographic groups. The original [FairFace paper](#) reports a total of 108,501 images annotated with age, gender, and race. However, for this assignment, students must use the public Hugging Face release of the dataset, which provides the publicly accessible train and validation portions required for model development and evaluation.

In the Hugging Face version, the dataset is provided in two configurations:

- 0.25
- 1.25

These two configurations represent different face-cropping settings:

- 0.25 contains more tightly cropped face images with less surrounding context and an image size of 224 x 224,
- 1.25 contains images with more surrounding visual context around the face and an image size of 448 x 448

For this assignment, students must use the 0.25 configuration only.

Within the 0.25 configuration, the publicly available Hugging Face dataset contains:

- **train:** 86,744 samples
- **validation:** 10,954 samples

This gives a total of **97,698 publicly accessible samples** for use in this assignment.

The dataset includes the following label fields:

- **age:** 9 age-group classes
- **gender:** 2 classes
- **race:** 7 classes
- **service_test:** an auxiliary boolean field that **may be ignored for this assignment**

The age label is the required prediction target for this assignment. The race and gender labels must be used for fairness auditing and subgroup performance analysis.

The age classes available in the dataset are:

- 0-2
- 3-9
- 10-19
- 20-29
- 30-39
- 40-49
- 50-59
- 60-69
- more than 70

The race classes available in the dataset are:

- East Asian
- Indian
- Black
- White
- Middle Eastern
- Latino_Hispanic
- Southeast Asian

The gender classes available in the dataset are:

- Male
- Female

Students must treat the Hugging Face validation split as the final held-out evaluation set. If an additional internal validation set is needed for model tuning, students should create it by splitting the official Hugging Face train split in a clearly documented and reproducible way.

Students can load the required FairFace dataset configuration from Hugging Face using the following code:

```
from datasets import load_dataset

dataset = load_dataset("HuggingFaceM4/FairFace", "0.25")
train_ds = dataset["train"]
test_ds = dataset["validation"]

print(train_ds)
print(test_ds)
print(train_ds.features["age"].names)
print(train_ds.features["gender"].names)
print(train_ds.features["race"].names)
```

7. Prediction Task

The required prediction task is age group classification. The model must take a single face image as input and predict one of the following age-group classes:

- 0-2
- 3-9
- 10-19
- 20-29
- 30-39
- 40-49
- 50-59
- 60-69
- 70+

The demographic labels used for fairness auditing must be:

- race
- gender

8. Minimum Project Requirements

Your project must satisfy all of the following:

- Use a CNN as the core learning model.
- Use FairFace as the primary dataset.
- Predict age group from a face image.

- Use a clearly documented train/validation/test split.
- Perform appropriate image preprocessing and at least some data augmentation.
- Train and evaluate at least one baseline CNN model.
- Report overall model performance using standard metrics.
- Evaluate performance across race and gender groups.
- Apply **at least two bias-mitigation strategies**.
- Compare baseline and mitigated models using evidence.
- Provide a final engineering recommendation on whether the prototype is acceptable for controlled deployment.

9. Required Technical Tasks

Task 1 - Problem Definition, and Exploratory Data Analysis

Define the prediction task as age group classification using the FairFace dataset and explain why this dataset is suitable for the assignment. Then perform an exploratory analysis of the dataset. At minimum, your analysis should include:

- number of samples,
- class distribution for age groups,
- demographic distribution across race and gender,
- examples of image variation,
- potential imbalance or underrepresentation,
- any missing, ambiguous, or noisy labels where applicable,
- a short discussion of dataset strengths and limitations.

Task 2 - Data Preparation

Prepare the dataset for CNN training. Document:

- train/validation/test split strategy,
- image resizing and normalization,
- standard data augmentation strategy,
- any cleaning or filtering performed,
- handling of labels and metadata used for prediction and fairness analysis.

Note: Deliberate bias-mitigation interventions such as rebalancing, oversampling, subgroup-balanced sampling, or class weighting should be introduced later during the mitigation stage.

Task 3 - Baseline CNN Model

Implement and train a baseline CNN model **from scratch using PyTorch** for age group classification on the FairFace dataset. You must design your own CNN architecture using standard layers such as convolution, activation, pooling, and fully connected layers. Clearly justify your architectural choices, including the number of layers, filter sizes, activation functions, and any regularization methods used.

Task 4 - Performance Evaluation

Evaluate the baseline model using appropriate metrics such as:

- accuracy,
- precision,
- recall,
- F1-score,
- confusion matrix,
- class-wise performance.

Task 5 - Bias Detection and Fairness Audit

Audit the model across race and gender groups. At minimum, include:

- group-wise accuracy or error rate across race groups,
- group-wise accuracy or error rate across gender groups,
- performance gap between best-performing and worst-performing groups,
- worst-group performance,
- short interpretation of what the gaps may indicate.

Task 6 - Bias Mitigation

Apply **at least two bias-mitigation strategies** to improve fairness in your CNN-based age group classification system. These strategies must be implemented without using pretrained models or transfer learning. Possible options include:

- oversampling of underrepresented groups,
- undersampling of overrepresented groups,
- class-weighted loss,
- balanced mini-batches,
- targeted augmentation,
- threshold adjustment where appropriate,

For each mitigation strategy, you must:

- explain why it was selected,
- describe how it was implemented,
- report the resulting overall performance and subgroup performance.

Task 7 - Comparative Analysis

Compare the baseline model with the mitigated model(s) and analyze:

- changes in overall performance,
- changes in subgroup performance across race and gender,
- fairness improvement versus performance trade-offs,
- technical benefits, limitations, and remaining risks.

Your conclusion should identify which model is most appropriate from both a technical and responsible AI perspective, and justify whether it is suitable for controlled deployment.

10. Project Constraints

To keep the project focused on CNN fundamentals, the following constraints apply:

- The core model must be CNN-based.
- The primary dataset must be FairFace.
- The required target variable must be age group.
- Race and gender must be used for fairness auditing.
- The CNN model must be designed and trained from scratch.
- The CNN model must be implemented and trained using **PyTorch**
- Pretrained models and transfer learning are not permitted.
- Extremely large-scale training is not required.

11. Technical Report Requirements

The report should be a concise technical report that documents the work completed for the required technical tasks. A long essay-style report is not expected. Students should focus on clearly presenting their methodology, results, analysis, and conclusions using appropriate visual evidence.

The recommended report length is **approximately 1500-2500 words**, excluding figures, tables, screenshots, references, and appendices.

The report must include clear supporting evidence where appropriate, such as:

- dataset distribution plots,
- sample images,
- training/validation curves,
- confusion matrices,
- subgroup performance tables,
- fairness comparison charts,
- screenshots of the demo application.

The report should include the following sections:

1. Title and student details
2. Problem definition and dataset overview
3. Exploratory data analysis
4. Data preparation and preprocessing
5. Baseline CNN model and training setup
6. Baseline evaluation results
7. Fairness audit across race and gender groups
8. Bias mitigation strategies and results
9. Comparative analysis of baseline and mitigated models
10. Demo overview with screenshots (the public URL of the hosted Streamlit application (see Section 13, 'Demo Requirement', for more information)
11. Ethical considerations, limitations, and final recommendation
12. References

Students are encouraged to present results in a clear and evidence-based manner using tables, graphs, and figures rather than unnecessary descriptive writing.

12. Technical Appendix (Jupyter Notebook)

Each student must submit **one main executed Jupyter Notebook file (.ipynb)** containing all technical work related to the required tasks in the assignment. The notebook should include dataset loading, exploratory analysis, preprocessing, baseline CNN development in PyTorch, training, evaluation, fairness analysis, mitigation experiments, and comparison of results. The notebook must be clearly organized, reproducible, and submitted with outputs visible, including relevant Markdown explanations, plots, and tables where appropriate.

13. Demo Requirement

Each student must provide a lightweight interactive demonstration of the developed system. The demo must be implemented as a simple web-based application, preferably using **Streamlit** (<https://streamlit.io/>), and hosted at a publicly accessible URL so that it can be reviewed remotely. The demo application should load and use the trained PyTorch model for inference and clearly show how the system works in practice.

At minimum, the demo should allow a user to upload or select a face image, display the top three predicted age-group classes, and show the confidence or probability associated with each prediction. The application should also include a brief description of the system and a short note on limitations and responsible use. The purpose of the demo is to communicate the practical usefulness of the trained model.

To support remote evaluation, each student must submit:

- the public URL of the hosted Streamlit application
- the source code and deployment files of the Streamlit-based web application as a single .zip file.

The hosted application must remain accessible until the grading period is complete. Real-time webcam processing, mobile deployment, large-scale backend systems, and production-grade deployment are optional and not required.

A short demo video of approximately 2-5 minutes.

The video should include:

- a brief high-level walkthrough of the main work completed in the Jupyter Notebook,
- a short explanation of the key stages of the project, such as dataset use, model training, evaluation, and bias-mitigation work,
- a demonstration of the hosted Streamlit application,

Additional requirements:

- the video must be submitted in **.mp4 format**
- the explanation must be given using the student's own natural voice,
- AI-generated voice narration is not permitted,
- the video must be presented in English,

14. Submission Package and Instructions

Each student must submit the following:

- Technical report in **.pdf** format. See **Section 11** for details.
- One executed Jupyter Notebook in **.ipynb** format with outputs visible. See **Section 12** for details.
- Source code and deployment files of the Streamlit-based web application as a single **.zip file**. See **Section 13** for details.
- Hosted Streamlit application URL for remote evaluation. See **Section 13** for details.
- Demo video in **.mp4 format**. See **Section 13** for details.

15. Assessment Criteria

Students will be assessed based on the overall quality, completeness, and correctness of their work. Evaluation will consider:

- understanding of the problem and dataset
- quality of exploratory data analysis and preprocessing
- correctness of CNN implementation in PyTorch
- quality of baseline training and evaluation
- depth of fairness analysis across race and gender groups
- quality and justification of bias-mitigation strategies
- clarity of comparison between baseline and mitigated models
- quality of the technical report, notebook, demo application, and video presentation

16. Academic Integrity and Originality Requirements

This is an applied engineering project. You may use open-source libraries, frameworks, and reference materials, but the final system design, experiments, analysis, and conclusions must be your own work.

The following are expected:

- clearly cite datasets, papers, tutorials, and reused code,
- explain all major design decisions in your own words,
- include original analysis rather than copying notebook outputs,
- discuss failures and limitations honestly.

Plagiarism, fabricated results, or uncredited copied work will be penalized according to institutional policy.

17. Final Note to Students

The strongest projects will not simply aim for the highest overall accuracy. They will demonstrate mature engineering judgment by balancing model performance, fairness, transparency, and responsible use considerations.